DataLab – Assignment 1 | Master IASD

Bayesian matrix factorization with feature-informed priors

Helping rarely rated movies borrow strength from their metadata

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Group 38 (LHOOQ) – October 2026

Idea 1: Bayesian averaging

43% of movies have fewer than 5 ratings: their latent factors cannot be pinned down.

ALS: one estimate



a single, possibly lucky, guess

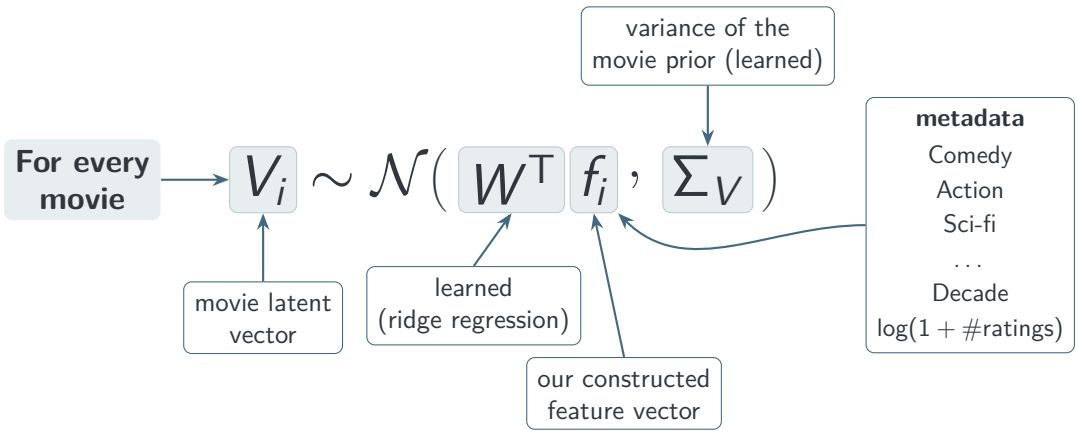
BPMF: many sampled estimates



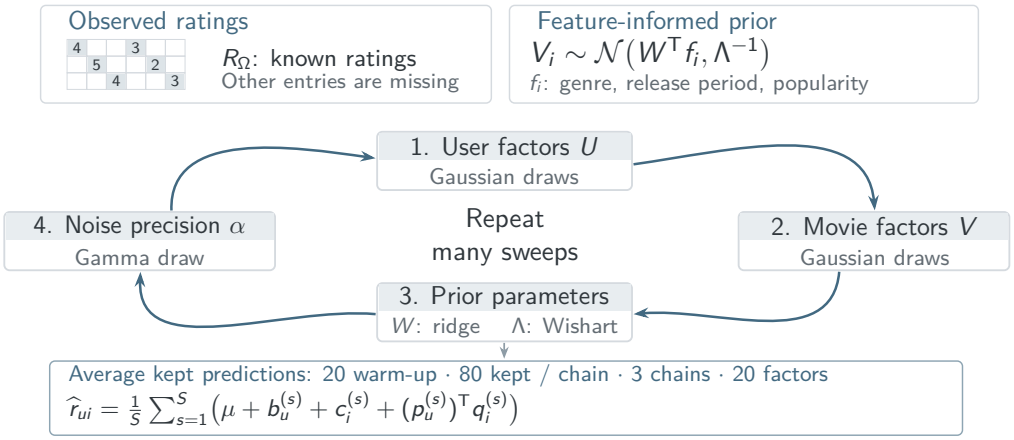
prediction = their average

$$\widehat{r}_{ui} = \frac{1}{S}\sum_{s=1}^{S}\left(\mu + b_u^{(s)} + c_i^{(s)} + p_u^{(s)\top} q_i^{(s)}\right)$$

New prior distribution



Fitting: Gibbs sampling



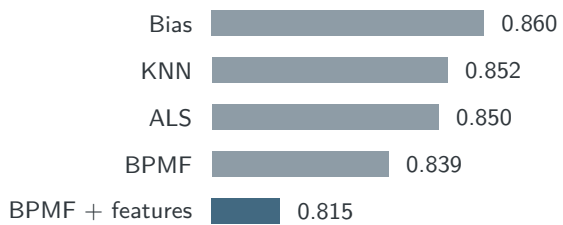
Results

Official platform (21 Oct)

0.806

RMSE · rank 8 · 121 s

Local comparison, same 12,639 held-out ratings



References

[1] **R. Salakhutdinov and A. Mnih** (ICML 2008)

Bayesian probabilistic matrix factorization using Markov chain Monte Carlo.

Used in slides 2 and 4: BPMF, Gibbs sampling, prediction averaging.

[2] **I. Porteous, A. Asuncion, and M. Welling** (AAAI 2010)

Bayesian matrix factorization with side information and Dirichlet process mixtures.

Related to slide 3: Bayesian MF using side information (different model).

[3] **R. P. Adams, G. E. Dahl, and I. Murray** (UAI 2010)

Incorporating side information in probabilistic matrix factorization with Gaussian processes.

Related to slide 3: GP-based side-information prior, not our linear prior.